Question 1 on Flexible AC Transmission System (FACTS)

a) No compensation

Steady-state stability limit

$$P_{max} = \frac{V_1 V_2}{X} = 2.5 \text{ pu}$$

[0.5 mark]

Initial power angle is

$$\delta_0 = \sin^{-1}\left(\frac{P}{P_{max}}\right) = 44.43°$$

[0.5 mark]

Critical clearing angle is obtained by equating the accelerating and decelerating areas:

$$\int_{\delta_0}^{\delta_{cr}} P \, d\delta = \int_{\delta_{cr}}^{\pi-\delta_0} (P_{max} \sin\delta - P) \, d\delta$$

$$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = \mathbf{66.46°}$$

[2 marks]

b) TCSC

With the TCSC the next steady-state stability limit would be

$$P_{max1} = \frac{V_1 V_2}{X(1-k)} = 3.33 \text{ pu}$$

[1 mark]

Initial power angle is

$$\delta_0 = \sin^{-1}\left(\frac{P}{P_{max1}}\right) = 31.67°$$

[1 mark]

Critical clearing angle is

$$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = \mathbf{77.42°}$$

[1 mark]

Model Answers

c) SVC

Steady-state stability limit with susceptance limit

$$P_{max2} = \frac{V_1 V_2}{X/2} = 5.0 \text{ pu}$$

Initial power angle δ_0 is given by:

$$\delta_0 = 2 \times \sin^{-1}\left(\frac{P}{P_{max2}}\right) = 40.97°$$

[1 mark]

$$B_{max} = 3.2 \text{ p.u.}$$

When the reactive capacity of the SVC is reached

$$\frac{4}{X}\left(1 - \cos\frac{\delta_{lim}}{2}\right) = V^2 B_{max} \rightarrow \delta_{lim} = 2 \times \cos^{-1}\left(1 - \frac{XB_{max}}{4}\right) = 94.31°$$

[1 mark]

Note $\delta_{lim} > 90°$ which dictates the following steps

[1 mark]

After the reactive capacity of the SVC is reached i.e. for $\delta > \delta_{lim}$ the power angle characteristics is given by

$$P = \frac{4}{X(4 - XB_{max})} \sin\delta = P_{max-lim} \sin\delta$$

$$P_{max-lim} = \frac{4}{0.4(4 - 0.4 \times 3.2)} = 3.68 \text{ p.u.}$$

[1 mark]

$$\delta_{max} = \pi - \sin^{-1}\left(\frac{P}{P_{max-lim}}\right) = 151.58°$$

[1 mark]

Equating the accelerating and decelerating areas:

$$\int_{\delta_0}^{\delta_{cr}} P \, d\delta = \int_{\delta_{cr}}^{\delta_{lim}} \left(P_{max2} \sin\frac{\delta}{2} - P\right) d\delta + \int_{\delta_{lim}}^{\delta_{max}} (P_{max-lim} \sin\delta - P) \, d\delta$$

[2 marks]

Model Answers

$$\delta_{cr} = 2 \times \cos^{-1}\left[\frac{P(\delta_{max} - \delta_0) + 2P_{max2}\cos\frac{\delta_{lim}}{2} - P_{max-lim}(\cos\delta_{lim} - \cos\delta_{max})}{2P_{max2}}\right] = \mathbf{87.54°}$$

[1 marks]

d) TCPAR

Ideal phase angle compensation does not change the steady-state stability limit, it would remain the same as the base case i.e., $P_{max} = 2.5$ pu

[1 mark]

With $\sigma_0 = -5°$, the initial power angle is given by

$P = P_{max}\sin(\delta + \sigma_0) \Rightarrow \delta_0 = \sin^{-1}\left(\frac{P}{P_{max}}\right) - \sigma_0 = 49.43°$

[1 mark]

Maximum power angle is given by

$$\delta_{max} = \pi - \sin^{-1}\left(\frac{P}{P_{max}}\right) + \sigma_{max} = 157.57°$$

[1 mark]

Critical clearing angle is obtained by equating the accelerating and decelerating areas:

$$\int_{\delta_0}^{\delta_{cr}} P\, d\delta = \int_{\delta_{cr}}^{\frac{\pi}{2}+\sigma_{max}} (P_{max} - P)\, d\delta + \int_{\frac{\pi}{2}+\sigma_{max}}^{\delta_{max}} [P_{max}\sin(\delta - \sigma_{max}) - P]\, d\delta$$

[1 mark]

$$\delta_{cr} = \left[\frac{\pi}{2} + \sigma_{max} - \cos(\delta_{max} - \sigma_{max})\right] - \frac{P}{P_{max}}(\delta_{max} - \delta_0) = \mathbf{77.22°}$$

[1 mark]

Model Answers

Question 2 on System Operation with Low Inertia

a)

i)

A plot of frequency revolution

[1 mark]

Rate of Change of Frequency (RoCoF), the instant immediately after the system disturbance
Frequency Nadir, the instant when the frequency deviation reaches a maximum value
Steady-state Frequency, the period after the frequency reaches the steady-state

[1 mark]

ii)

$2H\dot{f}(t) = R(t) - \Delta P_L,$
$R(t) = \frac{R}{T_d}t, \ \ if\ t < T_d,$
$\therefore f(t) = \frac{R}{4HT_d}t^2 - \frac{\Delta P_L}{2H}t,$

[1 mark]

$f_{nadir} = |\frac{\Delta P_L^2 T_d}{4HR}|,$

[1 mark]

$\therefore HR \geq \frac{\Delta P_L^2 T_d}{4 f_{max}}.$

[1 mark]

$R \geq \Delta P_L;$
$H \geq \frac{\Delta P_L}{2\ RoCoF_{max}}$

[1 mark]

iii)

$g(R, EFR) = (3 - 0.2EFR)R - 3(1.8 - EFR)^2,$
$\frac{\partial g}{\partial R} = 3 - 0.2EFR = 2.891,$
$\frac{\partial g}{\partial EFR} = -0.2R + 2(1.8 - EFR) = 7.203,$
$\frac{\partial R}{\partial EFR} = \frac{\frac{\partial g}{\partial EFR}}{\frac{\partial g}{\partial R}} = 2.49,$
$\frac{R}{EFR} = \frac{\partial R}{\partial EFR} \times \frac{c_{EFR}}{c_R} = 2.49 \times 1.6 = 4,$

[2 mark]

$3.8EFR^2 - 22.8EFR + 9.72 = 0,$
$EFR = 0.46$ GW,
$R = 4EFR = 1.84$ GW

[2 mark]

Model Answers

b)

i) BOOKWORK. The wind turbine can boost its power output by increasing the torque of the generator. This causes an imbalance of the torque that is applied to the rotating mass of the turbine, which causes the wind turbine to slow down. This causes the turbine to deviate from its optimal operating point, which causes a drop of its power coefficient, which aggravates the slow down of the turbine. As the turbine slows down, the torque has to be increased further, in order to sustain the power boost. This can't go on indefinitely because either the torque will exceed the rated torque of the generator or the speed will decrease beyond a lower threshold. These two variables will be monitored and the turbine will stop the power boost and start a recovery phase as soon as one of the two limits is breached.

[4 marks]

ii) This question can be solved using the swing equation, the angular acceleration divided by the inertia of a generator is equal to the imbalance between the mechanical torque of the prime mover and the electric torque exerted by the loads that draw power from the generator. Approximating for small variations of the speed and putting the inertia as a function of the ration between the kinetic energy of a plant and its rated power, this equation is of the following form:

$$\Delta P = -\frac{2H}{f_{nominal}} S_{base} \frac{df}{dt}$$

Students aren't expected to derive this equation on the spot, this is a well-known equation that has been covered in the lectures and it appears in undergraduate level textbooks on basics of electrical power systems. A reminder of this formula is also shown in part a).

Substituting ΔP, $f_{nominal}$ and df/dt into the equation, we can solve for H and it gives H=3.3 s.

[3 marks]

iii) The synthetic inertia service and the frequency droop services are defined as: $\Delta P_{inertia} = -\frac{2H}{f_{nominal}} S_{base} \frac{df}{dt}$ and $\Delta P_{droop} = -K(f - f_{droop})$. Looking at the graph of the frequency excursion, we can tell that point A corresponds to the frequency nadir, where df/dt is equal to zero. Consequently, only the frequency droop will affect the power output of the generator. Looking up the details of point A, we see that it's frequency is of 49.4 Hz, which means that the frequency error is of 0.6 Hz. Consequently, ΔP_{droop} =0.06 pu.

[3 marks]

Model Answers

Question 3 on Power System State Estimation

(i)

An operator needs to know:

- To what extent network is loaded
- If there any equality or inequality constraint violation
- If any overloads following planned or unplanned contingencies
- How best to control and reconfigure the network to utilise any spare capacity

[4 marks]

(ii)

Transmission System

- More measurements
  (Enough to guarantee observability)
- Methodologies and tools already exist
  (e.g. WLS, M-estimators, LAV)
- Pseudo measurements can be modelled as zero injection or inequality constraints
- Loads can be modelled as constant impedance type
- Limited variation in network parameters

Distribution System

- Less / inadequate measurements (observability not guaranteed)
- Lack of rigorous methodologies and tools for handling restricted measurements
- No definite model for pseudo measurements
  (Modelling pseudo measurements is challenging in DNO)
- High variability in network parameters (due to thermal variations etc)
- Loadings are unbalanced in nature

[4 marks]

(iii)

Set up:
State vector: node voltage and angle
Measurement:line real and reactive power flow, line current, bus voltage, power injection
Relate the measurement as function of state $z = h(x) + \varepsilon$
Error characterization: Gaussian, independent (uncorrelated), null average (unbiased) ,
standard deviation σ : $N(0,\sigma)$

Model Answers

$\text{cov}(\varepsilon) = \text{diag}(\sigma_i^2) = R$

Minimise a scalar function : $J(x) = [z - h(x)]^T W[z - h(x)]$;

$$J(x) = \sum_i W_i [z_i - h_i(x)]^2$$

Weight matrix : $W = R^{-1}$ Residual vector : $r = z - h(x)$

Solution:

First derivative = 0

$H^T W[z - h(x)] = 0$, $H = \partial h / \partial x$

Newton's approach

$G(x^k)\Delta x = H^t(x^k)W[z - h(x^k)]$

$G(x^k) = H^t(x^k)WH(x^k)$

G-gain matrix.

Obtain Δx and then update x^k . continue the iterative process until convergence

[5 marks]

(iv)

When the objective function of MLE has some constraints in the formulation we have equality constraints state estimation. Zero injection is one such measurement which is 100% correct, so best is to remove it from the measurement set rather than assigning a high weight. It reduces the dimensionality of the normal equation and along with constraint the solution is sought through Lagrange function. This is known as constraints state estimation. The problem set up is well conditioned for solution but symmetry is lost.

[3 marks]

(v)

When all measurements contributing to normal equation are of not equal quality, the normal equation is not well conditioned. In distribution system there are very few high-quality measurement, many pseudo measurements with large covariance, some leverage measurement – all needing highly varying weights. The normal equation is ill conditioned, requiring robust factorisation etc. The solution sometimes does not converge. Under such situation it is prudent to go for constraints state estimation approach

[2 marks]

(vi)

Model Answers

Critical measurement is the one that guarantees solvability and unique solution. It is also linked to observability. If critical measurement is lost, the whole power system is split into two sub systems and unified solution is not possible. The system is unobservable with the remaining set of measurements The critical measurement is also considered not to have residual error.

[2 marks]

Model Answers

Question 4 on Integrated Energy Systems

a)

i)

In the 1st hour,
Power from CCGT: 40000MW - 13000MW - 13000MW= 14000MW
Reserve requirement is 6000MW
Capacity required: 14000+6000=20000MW
20000/500=40 CCGTs minimum can be online.

[4 marks]

ii)

$$C = \min_{\{p_1 \dots p_{40}\}} 1 \cdot \sum_{i=1}^{40} (-0.2p_i + 160)p_i$$

$$\text{s.t.} \quad \sum_{i=1}^{40} p_i = 14000\text{MW}$$

$$300\text{MW} \le p_i \le 500\text{MW}, i \in \{1,2 \dots 40\}$$

$\Rightarrow$

$$\begin{cases} p_i = 500\text{MW}, i \in \{1,2 \dots 10\} \\ p_i = 300\text{MW}, i \in \{11,12 \dots 40\} \end{cases}$$

$$\boldsymbol{C = 1.2 £m}$$

[4 marks]

b)

i)

In a) where TES is not available,

in the second hour, 8000/200=40 CCGT is required to provide reserve,
minimum power from CCGT = 40*300=12000MW
Wind required = 40000-12000-13000 = 15000MW
Wind curtailment = 25000-15000=**10000MWh**

[2 marks]

In b) where TES is available,

in the 1st hour,
5000MW electrified heat load can be supplied by TES, so
Power from CCGT = 35000MW-13000MW-13000MW= 9000MW
Reserve from CCGT = 6000MW
Number of CCGT = (12000 + 6000)/500 = 36

[2 marks]

Model Answers

in the second hour, (8000)/200=40 CCGT is required to provide reserve,
minimum power from CCGT = 40*300=12000MW
Wind required = 40000-12000-13000 = 15000MW
The initial thermal energy stored in TES is 5000MWh, 5000MWh is consumed in the first hour, 0MWh is left in the second hour, considering the capacity is 7500MWh, 7500MW wind power can be stored in TES considering its energy capacity
Wind curtailment = (25000-15000-7500)*1h = 25**00MWh**
10000MW - 2500MWh = 7500MWh

[4 marks]

ii)

In b) where TES is available,

in the 1st hour,
Power from CCGT = 40000MW-13000MW-13000MW = 14000MW
Reserve from CCGT = 1000MW
Number of CCGT = (14000 + 1000)/500 = 30

[2 marks]

in the second hour, (8000-5000)/200=15 CCGT is required to provide reserve,
minimum power from CCGT = 15*300=4500MW
Wind required = 40000-4500-13000 = 22500MW
The initial thermal energy stored in TES is 5000MWh, 0MWh is consumed in the first hour, 5000MWh is left in the second hour, considering the capacity is 7500MWh, 2500MW wind power can be stored in TES considering its energy capacity
Wind curtailment = (25000-22500-2500)*1h = **0MWh**

[2 marks]

Model Answers